

1. Objectives:

1. To understand the basic structure of a Java program including package, class, and main method.
2. To implement user input and output operations using the Scanner class in java.
3. To apply fundamental programming concepts such as variables, data types, and console interaction.

2. Introduction:

Java is a high-level, object-oriented programming language originally developed by Sun Microsystems and later maintained by Oracle Corporation. It is platform-independent, meaning that Java programs can run on any system that supports the Java Virtual Machine (JVM). Every Java program is written inside a class, and execution begins from the public static void main(String[] args) method. The use of packages helps organize related classes and maintain a structured project environment.

In Java, user input is taken using the Scanner class from the java.util package. The Scanner class is a predefined class that allows a program to read different types of data such as integers, floating-point numbers, and strings from input sources like the keyboard. Methods such as nextInt(), nextFloat(), and next() are used depending on the required data type. Output is displayed using System.out.print() and System.out.println(). Java also supports escape sequences such as \n (new line), \t (tab space), \" (double quotation mark), and \\ (backslash), which help in formatting the output properly.

In addition, Java provides formatted output using format specifiers with methods like System.out.printf(). A format specifier is a placeholder that defines how a value should be displayed in the output. For example, %d is used for integers, %f for floating-point numbers, %s for strings, and %.2f for displaying a floating-point number up to two decimal places. Various operators are also used in Java to perform computations and logical decisions. Arithmetic operators (+, -, *, /, %) are used for calculations, relational operators (>, <, ==, !=) are used for comparison, and logical operators (&&, ||) are used to combine multiple conditions. These features collectively help in controlling program flow and solving real-world problems effectively.

3. Required Hardware and Software:

- **Hardware:** AMD RYZEN 7, RAM 8 GB
- **Software:** Visual Studio Code 1.109

4. Experiment Implementation:

Experiment 1: User input using Scanner Class

```
package SecondLab;

import java.util.Scanner;

public class ClassOne {

    public static void main(String[] args) {

        Scanner ac = new Scanner(System.in);

        System.out.print("Enter a Number: ");
        int a = ac.nextInt();
        System.out.println("The Number is: " + a);
        System.out.println();
    }
}
```

```

        System.out.print("Enter a Letter: ");
        String b = ac.next();
        System.out.println("The letter is: " + b);
        System.out.println();

        ac.close();

    }
}

```

2.1: Experiment 1 output

```

PS D:\BAUST CSE\Java Programing\2nd_Lab> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe'
'-cp' 'C:\Users\Rezwan\AppData\Roaming\Code\User\workspaceStorage\01e60a4cf252fd32ce3a59c
b.ClassOne'
Enter a Number: 5
The Number is: 5

Enter a Letter: R
The letter is: R

PS D:\BAUST CSE\Java Programing\2nd_Lab>

```

Experiment 2: Parameter of Rectangle

```

package SecondLab;

import java.util.Scanner;

public class ClassTwo {

    public static void main(String[] args) {

        Scanner cc = new Scanner(System.in);
        System.out.print("Enter 1st Float Number: ");
        float c = cc.nextFloat();

        Scanner dc = new Scanner(System.in);
        System.out.print("Enter 2nd Float Number: ");
        float d = cc.nextFloat();

        float Number = 2 * (c + d);
        System.out.println("The Answer is " + Number);

        cc.close();

    }
}

```

2.2: Experiment 2 output

```

PS D:\BAUST CSE\Java Programing\2nd_Lab> & 'C:\Program Files\Java\jdk-25
sages' '-cp' 'C:\Users\Rezwan\AppData\Roaming\Code\User\workspaceStorage\0
'SecondLab.ClassTwo'
Enter 1st Float Number: 1.5
Enter 2nd Float Number: 2.5
The Answer is 8.0
PS D:\BAUST CSE\Java Programing\2nd_Lab>

```

Experiment 3: Number Comparison

```

package SecondLab;

import java.util.Scanner;

public class ClassThree {

    public static void main(String[] args) {

        Scanner ec = new Scanner(System.in);
        System.out.print("Enter 1st Number: ");

```

```

        int e = ec.nextInt();

        Scanner ec = new Scanner(System.in);
        System.out.print("Enter 2nd Number: ");
        int f = ec.nextInt();

        if (e > f) {
            System.out.println("1st Number is Greater");
        } else if (f > e) {
            System.out.println("2nd Number is Greater");
        } else {
            System.out.println("Both Numbers are Equal");
        }

        ec.close();
    }
}

```

2.3: Experiment 3 output

```

● PS D:\BAUST CSE\Java Programing\2nd_Lab> & 'C:\Program Files\Java\jdk-
sages' '-cp' 'C:\Users\Rezwan\AppData\Roaming\Code\User\workspaceStorag
'SecondLab.ClassThree'
Enter 1st Number: 5
Enter 2nd Number: 10
2nd Number is Greater
○ PS D:\BAUST CSE\Java Programing\2nd_Lab>

```

Experiment 4: Even or Odd

```

package SecondLab;

import java.util.Scanner;

public class ClassFour {

    public static void main(String[] args) {

        Scanner gc = new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int g = gc.nextInt();

        if (g % 2 == 0) {
            System.out.println("The Number is Even");
        } else {
            System.out.println("The Number is Odd");
        }

        gc.close();
    }
}

```

2.4: Experiment 4 output

```

● PS D:\BAUST CSE\Java Programing\2nd_Lab> & 'C:\Program Files\Java\jdk-
sages' '-cp' 'C:\Users\Rezwan\AppData\Roaming\Code\User\workspaceStora
'SecondLab.ClassFour'
Enter a Number: 10
The Number is Even
○ PS D:\BAUST CSE\Java Programing\2nd_Lab>

```

Experiment 5: Leap Year

```
package SecondLab;

import java.util.Scanner;

public class ClassFive {

    public static void main(String[] args) {

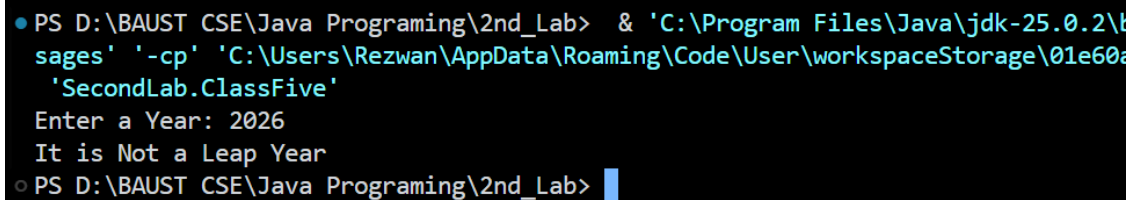
        Scanner hc = new Scanner(System.in);
        System.out.print("Enter a Year: ");
        int year = hc.nextInt();

        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
            System.out.println("It is a Leap Year");
        } else {
            System.out.println("It is Not a Leap Year");
        }

        hc.close();
    }

}
```

2.5: Experiment 5 output



```
PS D:\BAUST CSE\Java Programing\2nd_Lab> java -cp 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' 'C:\Users\Rezwan\AppData\Roaming\Code\User\workspaceStorage\01e60a\SecondLab\ClassFive'
Enter a Year: 2026
It is Not a Leap Year
PS D:\BAUST CSE\Java Programing\2nd_Lab>
```

5. Discussion & Conclusion:

In this lab, we have learned the fundamental concepts of Java programming. The basic structure of a Java program, including package declaration, class definition, and the main() method, was clearly understood. The use of the Scanner class for taking user input, different data types such as int, float, and String, and the role of escape sequences in formatting output were studied. Various operators including arithmetic, relational, and logical operators were also practiced to understand how decision-making and calculations are performed in Java. After understanding these concepts, we took user input and displaying output, performing arithmetic calculations, comparing two numbers, checking whether a number is even or odd, and determining whether a year is a leap year. Through these implementations, theoretical knowledge was successfully transformed into practical programming skills, which build a strong foundation for advanced Java development.

6. Reference:

1. <https://www.geeksforgeeks.org/java/scanner-class-in-java/> accessed at 6.25 PM on 4th March 2026.
2. <https://www.geeksforgeeks.org/java/escape-sequences-in-java/> accessed at 8.05 PM on 4th March 2026.